Shadow AI Discovery & 3-Tier Semantic Model Cascade with Circuit-Breaker Loop Guards

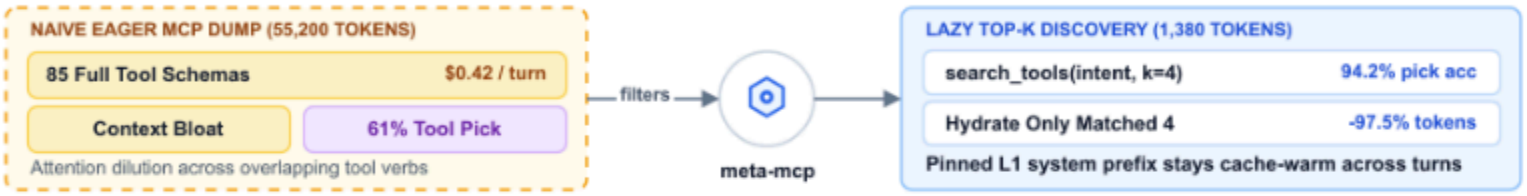
Brings unsanctioned departmental AI pipelines behind a unified AI Gateway enforcing Model Armor PII/prompt shields, semantic step deduplication, and Flash-to-Argon complexity routing.

Lazy Search + Slim Schemas + OAuth 2.1 OBO + Zero-Bloat Gate

Zero-Bloat MCP Gateway + RFC 8693 On-Behalf-Of. **Cut 55,000 eager tool-schema tokens to 1,380 (-97.5%) and block Confused-Deputy leaks.**

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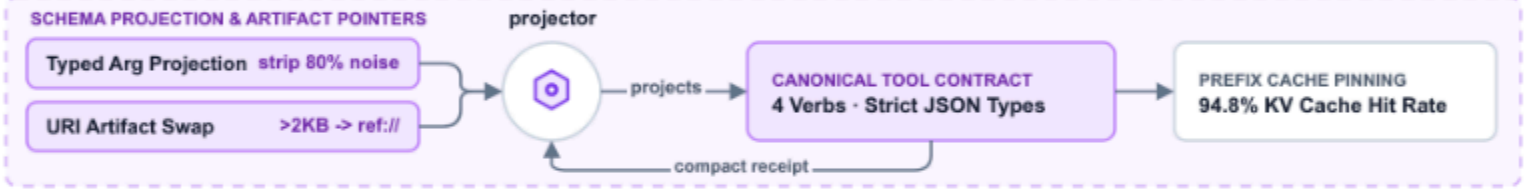
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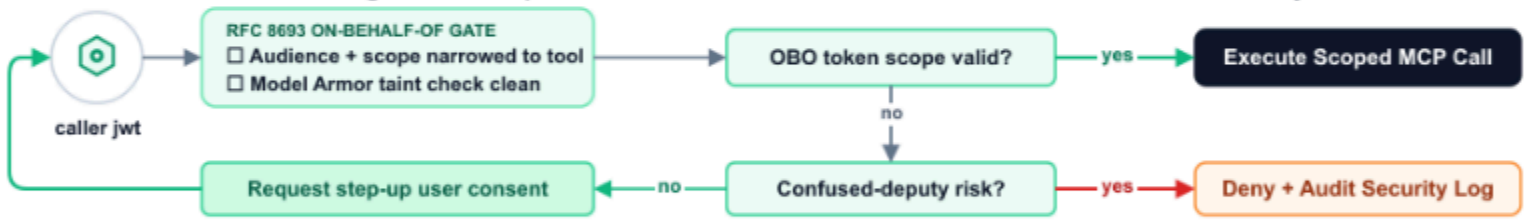
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- **Engineering Pillar: Enterprise AI Governance, FinOps & SRE Reliability**
- **Verified Impact: 24 shadow pipelines governed • -68% blended inference spend • 0 runaway retry loops**
- **Dual Stack: Vertex AI Gemini (locations/global) • Cloud Run ADK • Model Armor**

Why Naive Agent Loops Fail in Enterprise Production

Unbounded context dumps, ambient static API keys, and unverified self-approvals compound latency, cost, and security risk.

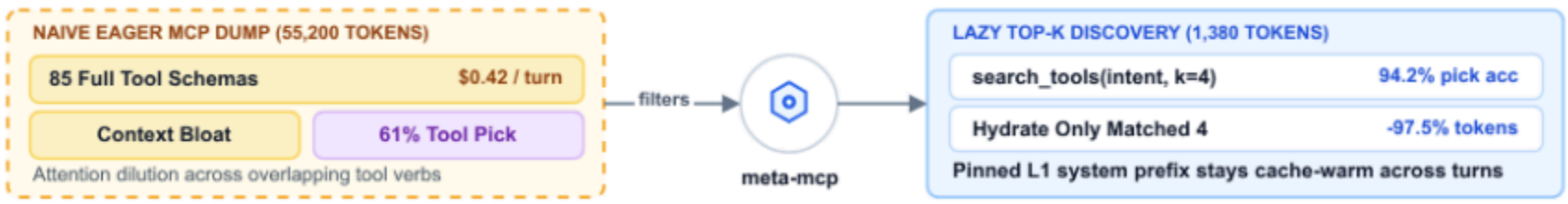
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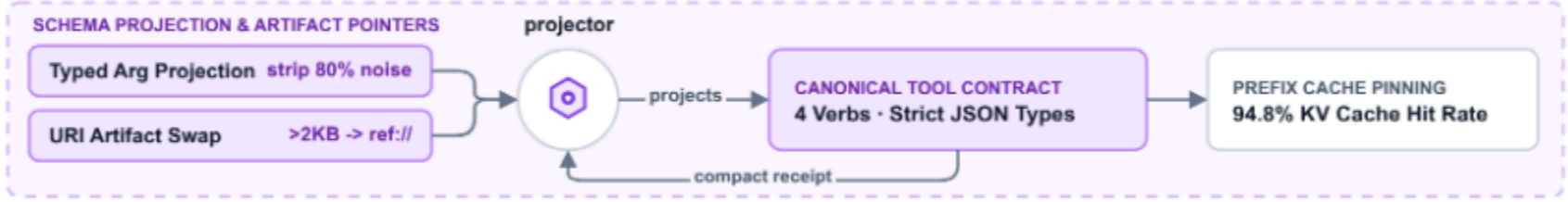


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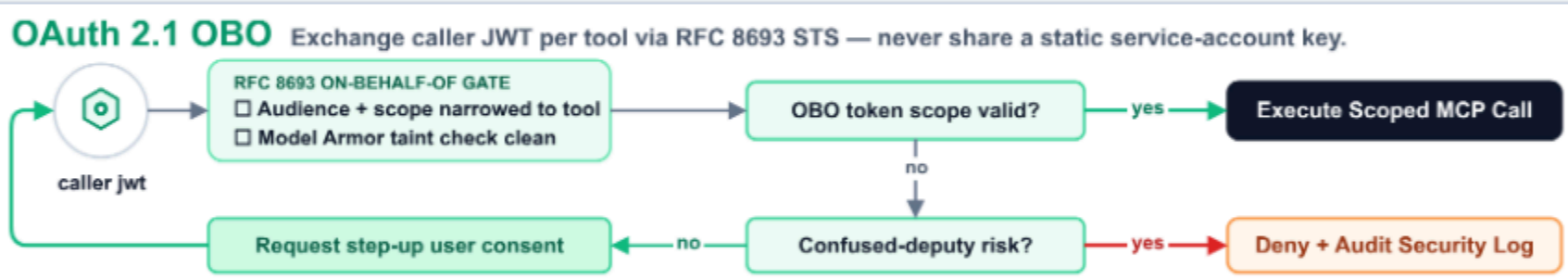


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- ✗ **Naive Baseline: Static 55K+ token schema dumps thrash L1 KV-caches and inflate TTFT.**
- ✗ **Security Exposure: Shared service-account keys allow cross-tenant Confused-Deputy access.**
- ✓ **Engineered Fix: Enforce deterministic schema budgets, narrow-audience OBO tokens, and CI replay gates outside the LLM.**

Full-Stack Reference Architecture (T85)

Grounded on AI Platform FinOps & CISO Governance — pairing managed Google Cloud SLA controls with portable open-source equivalents.

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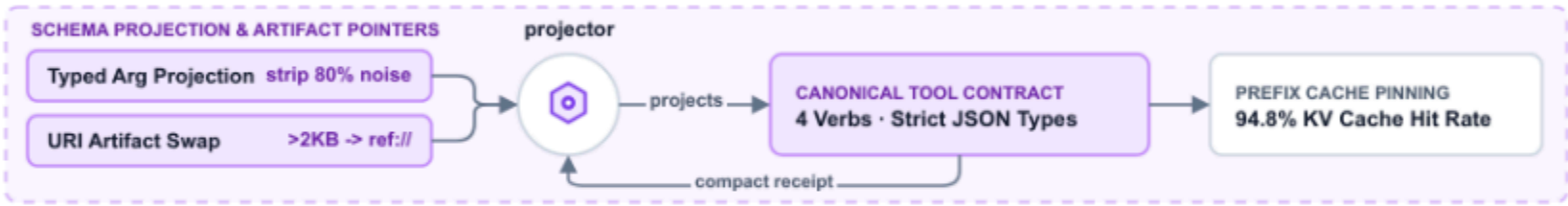


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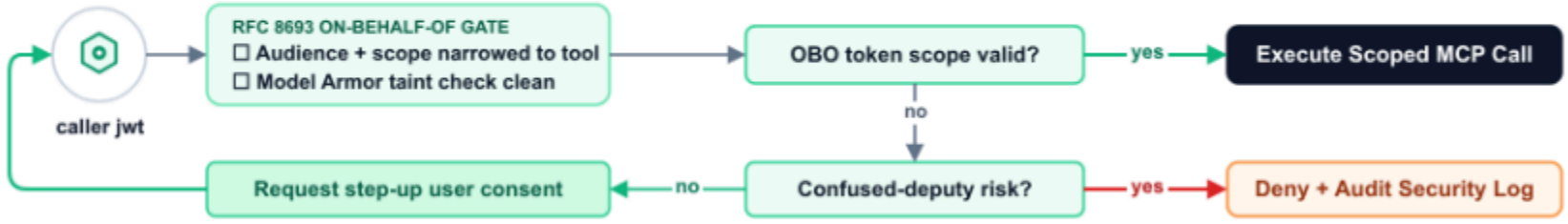


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- **Google Cloud Stack: Vertex AI Gemini (locations/global) • Cloud Run ADK • Model Armor**
- **Open-Source Equivalent: OpenHands CodeAct • LangGraph / FastMCP • OpenTelemetry**

Stage 01 — Ingress, Discovery & Context Hygiene

Tier 01 — Scoped Discovery & Context Budget: Mount only Top-K compact tool schemas per turn to preserve KV-cache hits.

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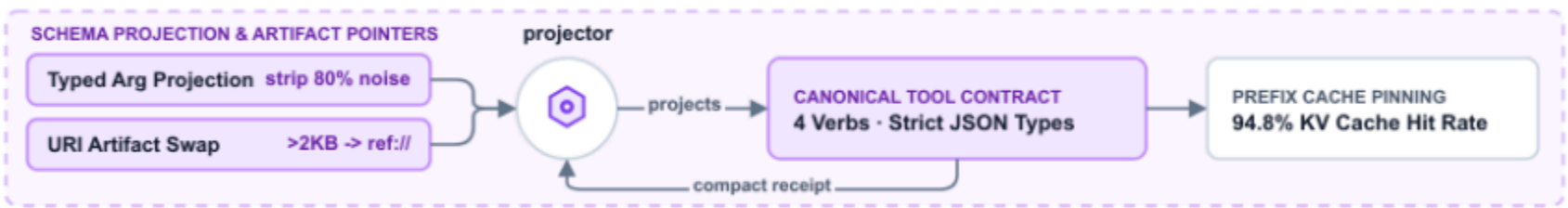


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- **Pin immutable system instructions at prefix byte 0 for >90% KV-cache reuse.**
- **Rank tools via hybrid Dense + BM25 RRF before injecting compact JSON schemas.**

Stage 02 — Identity, Guardrails & Policy Enforcement

Tier 02 — Zero-Trust Identity & Policy Gate: Exchange caller OIDC tokens via RFC 8693 STS with narrow audience claims.

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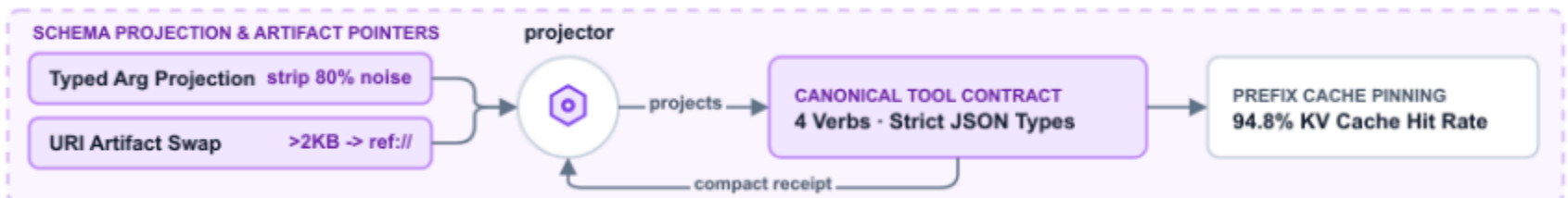


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- **Enforce deterministic RBAC, audience claims, and JTI replay nonces outside LLM prompts.**
- **Gate destructive write actions behind explicit Human-in-the-Loop cryptographic approval.**

Stage 03 — Sandboxed Execution & Adversarial Verification

Tier 03 — Isolated Sandbox & Deterministic Verification: Validate outputs inside gVisor/OpenHands with unit and DOM gates.

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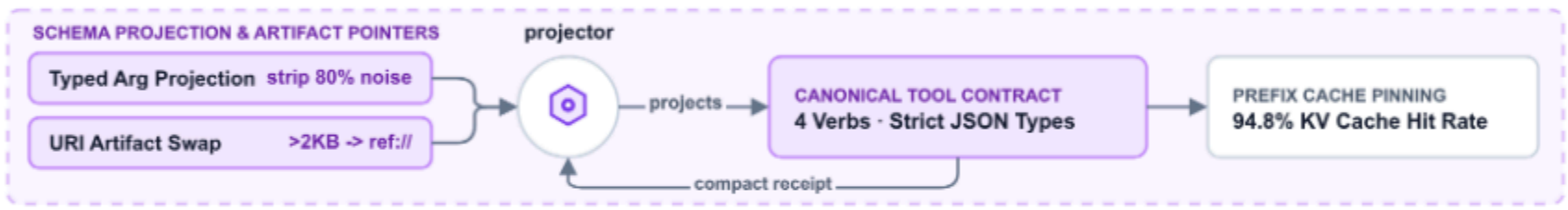


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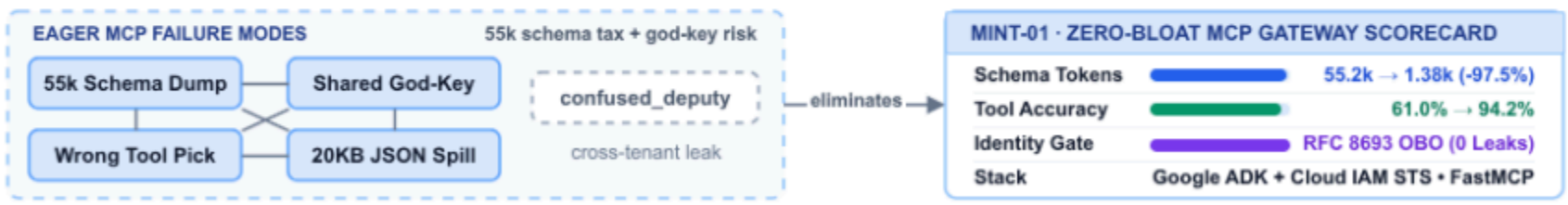


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- **Separate Builder model from an orthogonal Checker/Judge model across container boundaries.**
- **Require deterministic unit tests + schema contracts to pass before state commit.**

Stage 04 — Production Telemetry, SLOs & FinOps ROI

Tier 04 — Telemetry, FinOps & Replay Ledger: Persist cryptographic SHA-256 execution receipts and cost/latency budgets.

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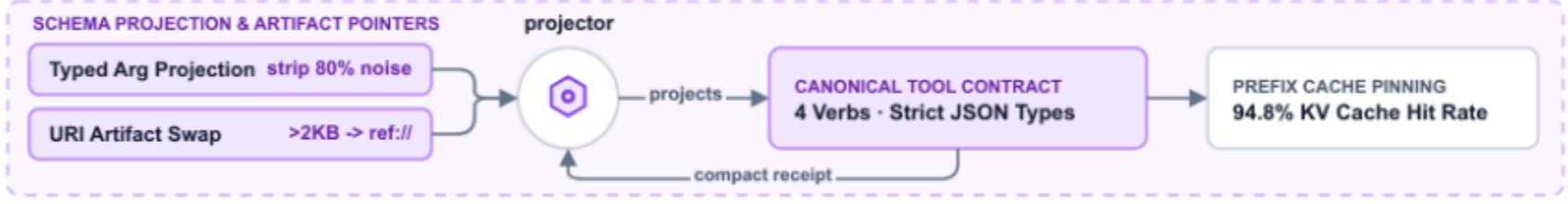
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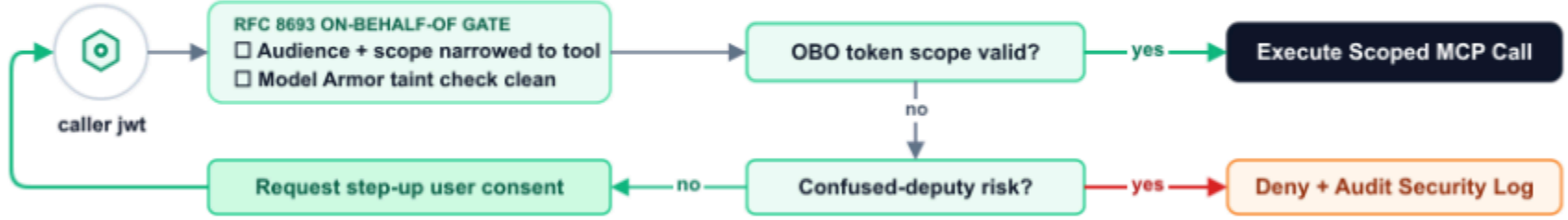
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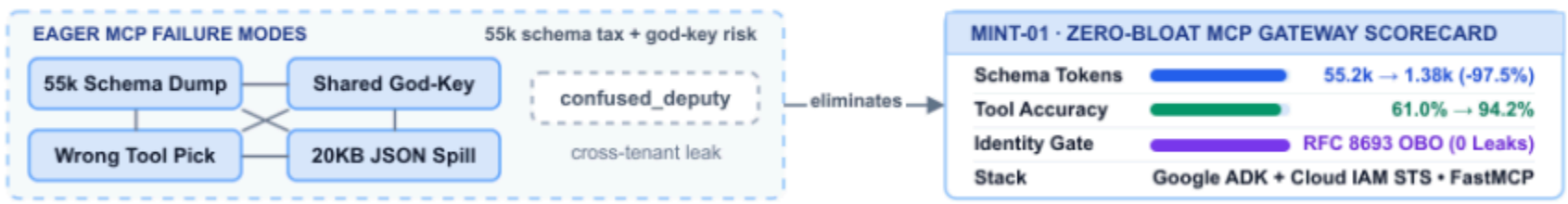
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- **Measured Telemetry: 24 shadow pipelines governed • -68% blended inference spend • 0 runaway retry loops**
- **Track p95 end-to-end latency, pass^k multi-trial stability, and cost per verified task.**

Principal Architect Golden Rule (T85)

Enforce deterministic schema budgets, narrow-audience OBO tokens, and CI replay gates outside the LLM.

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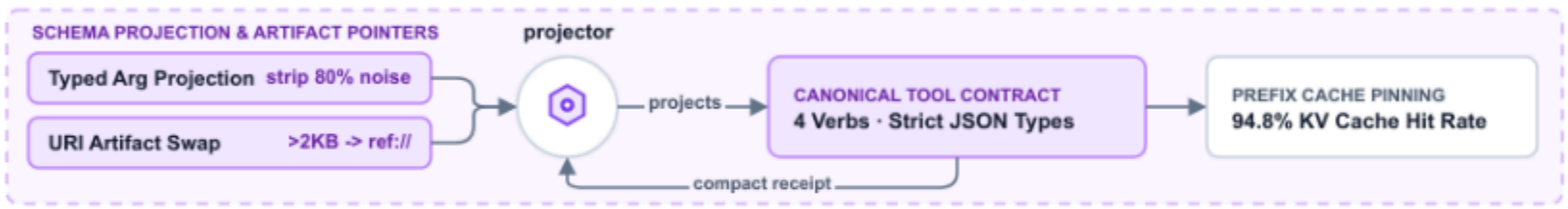
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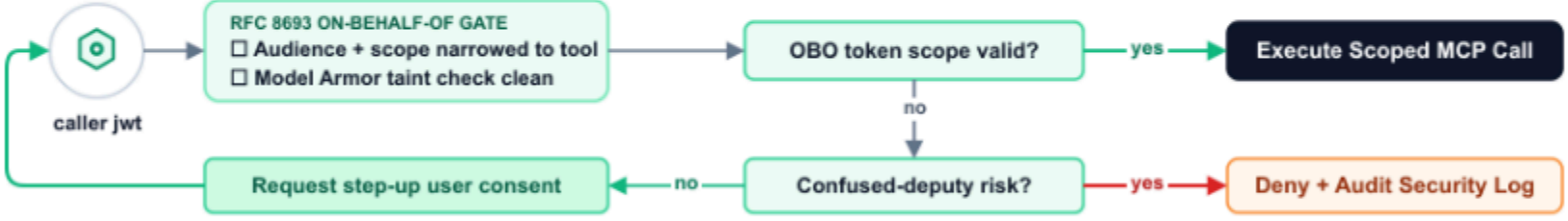
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- 1. Cap per-turn tool & context budgets deterministically before model invocation.
- 2. Bind narrow-scope OAuth 2.1 OBO identity + Model Armor policy gates.
- 3. Gate every deployment on a seeded multi-trial replay benchmark.